

ENCAPSULATION



A capsule type tablet is used to store/bind the medicine, which is in powdered or granular form. It uses a silicon coating to bind them together.

Definition:

The process of binding the attributes and behaviour of an object using a class is called as Encapsulation.

Purpose of Encapsulation:

DATA HIDING – The process of restricting the direct access to data members and allowing only indirect access to data members via methods. Encapsulation is restricted to attributes and not applicable to methods.

Advantages of Data Hiding:

1. Increased privacy/security
2. Making data members only hidden
3. Making data members only readable
4. Making data members only writable
5. Making data members both readable and writable

Methods used in Encapsulation:

Two non-static methods called `getter()` and `setter()` are used. They have pre-defined way to declare.

Getter():

```
access_modifier return_type get_attribute_name()
{
    return attribute_name;
}
```

Example for Getter method:

```
2
3 class Laptop
4 {
5     String brand;
6     //GETTER METHOD
7     public String getBrand()
8     {
9         return brand;
10    }
11 }
```

Characteristics of Getter method:

1. It is a no argument method.
2. It returns the data of the attribute.
3. It has return type as the datatype of the attribute that is returned.
4. It is used to read/access/print the data of the attribute.

Setter():

```
access_modifier void set_attribute_name(datatype attribute_name)
{
    this.attribute_name = attribute_name;
}
```

Example for Setter method:

```
class Laptop
{
    String brand;
    //SETTER METHOD
    public void setBrand(String brand)
    {
        this.brand = brand;
    }
}
```

Characteristics of Setter method:

1. It is a parameterized method.
2. It is used to re-initialize the attribute with the user passed value.
3. The return type of the setter method is void.
4. It is used to write/re-initialize/modify the data of the attribute.

UML diagram representation for Encapsulation:

MARKER

-brand
-colour (r)
-price (w)
-type (r/w)

(-) sign indicates that only hidden, by private modifier
(r) sign means only readable, only getter method
(w) sign means only writable, only setter method
(r/w) sign means both readable and writable, usage of both getter and setter methods

Example program for Encapsulation:

```
1 package javapackage;
2
3 class EncapsulationDriver
4 {
5     public static void main(String[] args)
6     {
7         AirConditioner ac = new AirConditioner();
8         System.out.println("*****AC DETAILS*****");
9         System.out.println("brand of AC is: "+ac.getBrand());
10        ac.setBrand("LLOYD");
11        ac.setBrand("VOLTAS");
12        System.out.println("modified brand of AC is: "+ac.getBrand());
13        System.out.println("color of AC is: "+ac.getColor());
14        System.out.println("price of AC is:"+ac.getPrice());
15        ac.setPrice(47652.24f);
16        System.out.println("modified price of AC is:"+ac.getPrice());
17        System.out.println("capacity of AC is: "+ac.capacity);
18    }
19 }
20
21 class AirConditioner
22 {
23     private String brand;
24     private String color;
25     private float price;
26     //private float star;
27     //star is not used
28     int capacity;
29     public String getBrand()
30     {
31         return brand;
32     }
33     public void setBrand(String brand)
34     {
35         this.brand = brand;
36     }
37     public String getColor()
38     {
39         return color;
40     }
41     public float getPrice()
42     {
43         return price;
44     }
45     public void setPrice(float price)
46     {
47         this.price = price;
48     }
49 }
50
51
```

<terminated> EncapsulationDriver [Java Applet]
*****AC DETAILS*****
brand of AC is: null
modified brand of AC is: VOLTAS
color of AC is: null
price of AC is:0.0
modified price of AC is:47652.24
capacity of AC is: 0